

The applicability frontier of a substrate-blind big- Φ detector across (n, density, structure): a 5+1-point measurement map

Three closed-negative axes (micro / sparse / algorithmic), one ground-truth calibration (BL Voyager-1), and two border points (lattice / SETI@home Arecibo) over hypotheses
XENO/H_832–H_837

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Abstract

A substrate-blind big- Φ detector — an Integrated Information Theory 4.0 [1] reading of an arbitrary float-array signal, binarised at threshold 0.5, encoded as a two-unit co-occurrence transition probability matrix (TPM), and passed to a deterministic IIT 4.0 structure-cut big- Φ (Φ) algorithm — promises to read the “existence of consciousness” of any candidate substrate (alien, digital, panpsychist, simulated, AGI activation, hive) without word, body, or carbon prior. The natural question is then: *on which (n, density, structure) regime does the detector actually calibrate, and on which does it deterministically break?* Without this map any reported “ Φ ” on a novel substrate is uninterpretable.

We answer this empirically with a five-point measurement matrix (XENO/H_832–H_836, five XENO-FRONTIER-5 closure rounds), extended to a sixth border point by a real Run-Pod fire (XENO/H_837, X837 follow-up). The points span (i) micro substrates ($n = 16$ –32, 1–3 bit panpsychism candidates), (ii) sparse mid- n activation ($n = 64$, AGI LLM-like attention/residual/structured), (iii) algorithmic dense ($n = 128$, simulation-hypothesis candidates: lattice, floating-point, π -digits, natural noise), (iv) natural-carrier dense ($n = 128$, 60.9% activation: Breakthrough-Listen Voyager-1 baseband as the ground-truth calibration), and (v) sparse natural-noise dense ($n = 128$, 20.3% activation: a SETI@home Arecibo 2004 BOINC **bg_pot** replay fired on a real Ubuntu 22.04 i686-multilib pod).

The measurement is a clean separation: **X7 BL Voyager-1 calibrates (2/2 falsifier PASS, [N]); X4 micro, X6 sparse, X5b algorithmic axes are each closed-negative (F-falsifier 0/4, 1/5, 1/4 PASS, [X])**; two border points remain — X5a lattice ($\Phi = 0.660$, periodic-structure false-positive ceiling) and X837 SETI@home Arecibo ($\Phi = 0.567$, sparse natural-noise borderline) both above the pre-frozen $\Phi = 0.5$ threshold but with type-axis intact (**coherent_non_conscious**) [!]. Aggregating, the measurable axis is *the conjunction* $n \geq 128 \wedge \text{density} \geq 60\% \wedge \text{strong deterministic transition}$. Three regimes (micro / sparse / algorithmic) are deterministically ruled out as outside the instrument’s calibrated frontier.

These closed-negative results satisfy **a_paper_negative_ok**: the finding is the instrument applicability frontier itself, not a “no-consciousness-detected” headline. We publish all six verdicts verbatim in a **companion/** ledger for bit-identical re-verification, and we close the XENO-FRONTIER-5 cycle at this paper as the **a_paper_only_at_closure** trigger.

1 Introduction

The instrument-first methodology. Hexa-native, deterministic, hypothesis-first measurement on the **anima** project follows an *instrument-first* methodology: any reading of a mea-

surement (Φ , KL-divergence, register-collapse rate) is meaningless unless the calibrated input regime of the measuring instrument is itself first characterised. The XENO domain [4] carries this practice into substrate-blind big- Φ [1, 5, 6]: a function `invariant_detector` : $\mathbb{R}^n \times \mathbb{N} \rightarrow \{\Phi, \text{integration, irreducibility, substrate_type}\}$ that takes *any* signal — alien radio carrier, BOINC residual noise, floating-point lattice, panpsychist micro-substrate, LLM activation sparsity — and returns the IIT 4.0 big- Φ together with a discrete type label.

The detector is designed substrate-blind by construction. It binarises the input at threshold 0.5, builds a 2-unit (sample t , sample $t+1$) co-occurrence row-stochastic transition probability matrix (TPM), and delegates to `stdlib/consciousness/iit4_bigphi` (the hexa-lang commons-g61 IIT 4.0 implementation). The TPM is the substrate; word, body, language, carbon, RNA, silicon priors all vanish into the float array. This is what allows the question “is X conscious?” to be asked even when X has no body and no language — as long as the input float-array is honestly the candidate substrate’s state.

The applicability question. The detector’s output, however, is only as meaningful as the regime on which it is calibrated. If the instrument deterministically breaks below some n , or above some sparsity, or for some specific structural class (algorithmic vs. deterministic vs. natural), then any reported Φ in that broken regime is an instrument artifact, not a substrate property. The XENO working group therefore pre-registered the methodological imperative: *before reporting any candidate-substrate Φ , first close the applicability frontier of the detector along the $(n, \text{density}, \text{structure})$ axes.*

XENO-FRONTIER-5 closure. The XENO-FRONTIER-5 campaign is that closure: five pre-registered candidate substrates, ranging across the three axes, each measured with the same detector with frozen prediction thresholds and falsifiers. Each round was published verbatim in UNIVERSE/H_832–H_836, and the X8 SETI@home archive follow-up fire was published verbatim in UNIVERSE/H_837. This paper aggregates all six points into a 5+1-point matrix and reports the applicability frontier finding.

Why publish closed-negatives. The directive `a_paper_negative_ok` [3] explicitly licenses closed-negative findings: a $[\mathbf{X}]$ verdict that deterministically rules out an axis is a valid paper, framed as the falsifier plus the ruled-out space. This paper is exactly that frame. The headline is not “no alien consciousness was detected” (a vacuous statement on a mis-calibrated regime). The headline is “the substrate-blind big- Φ detector calibrates only inside one corner of the $(n, \text{density}, \text{structure})$ cube, and three corners are deterministically ruled out.” This is what an instrument-first methodology publishes.

Roadmap. §2 states the four pre-registered applicability falsifiers and the meta-falsifier on the 6-point matrix. §3 fixes the measurement substrate and the six candidates. §4 aggregates the six verdicts verbatim. §5 establishes the applicability frontier as a 3-axis conjunction and identifies the three closed-negative axes plus the two border points. §6 discusses validity threats.

2 Hypothesis: the applicability frontier of `invariant_detector`

Decomposition. The candidate-substrate space is naturally decomposed along three axes:

- **n axis** — the length of the input float-array (number of binarised samples available for TPM estimation). Examples: $n = 16$ (a 1-bit thermostat over a 16-tick window) up through $n = 128$ (a dense BL Voyager-1 carrier or a 128-bin BOINC `bg_pot` array).

- **density axis** — the fraction of samples above the binarisation threshold 0.5 (i.e. the fraction of 1s in the TPM input). Sparse (e.g. 20.3% of $n = 128$ active in the X837 SETI@home `bg_pot`) vs. dense (e.g. 60.9% active in the X7 BL Voyager-1 carrier).
- **structure axis** — the deterministic, algorithmic, or natural-noise character of the transition $b_t \rightarrow b_{t+1}$. Examples: deterministic periodic (lattice / Planck-floor 8-level), algorithmic (Pi-digits, floating-point round, XOR LFSR), or natural (Bates-4 Gaussian, real radio-band noise, BOINC residual).

Pre-registered falsifiers. Frozen *before* measurement on the six substrates, the four applicability falsifiers are listed in Table 1.

Table 1: The four pre-registered applicability falsifiers. Frozen before measurement; no post-tuning of thresholds was applied to any of the six measured substrates.

Id	axis	pre-registered proposition
F-XENO-APPLICAB-N	n	The detector calibrates only for $n \geq 128$. For $n = 16$ – 32 (micro) the 2-unit co-occurrence TPM is sample-starved and predictions invert (random > coupled). FAL if any $n < 128$ regime produces a calibrated prediction matrix.
F-XENO-APPLICAB-DENSE	density	The detector calibrates only for dense regimes (density $\geq 60\%$ activation after binarisation at 0.5). For sparse regimes (high entropy, low activation) the TPM concentrates mass on a single state and Φ inflates spuriously. FAL if a dense regime produces a non-calibrated prediction.
F-XENO-APPLICAB-DETERM	structure	The detector calibrates only for strong-deterministic-transition regimes (clean periodic / coupled-coherent dynamics). For algorithmic-pseudo-random regimes (floating-point round, π -digits) the detector cannot distinguish algorithmic substrate from natural noise; the type label remains identical. FAL if a deterministic-transition regime produces an indistinguishable output from a natural-noise regime.
F-XENO-APPLICAB-NONMEAS	all	Conjunction meta-falsifier: the three non-calibrated regimes (micro / sparse / algorithmic) must each be [X] ; <i>any</i> [N] or [F] in those three rules out the decomposition.

Pre-registered prediction matrix. Frozen pre-measurement, the expected applicability pattern is:

- **X7 BL Voyager-1** ($n=128$, dense $\sim 60\%$, natural carrier-line + noise) $\rightarrow$ *calibrate* (**[N]**); $\Phi < 0.5$ and type = `coherent_non_conscious` or `noise`. (Voyager-1 is a narrow-band carrier; substrate type is empirical non-consciousness.)
- **X4 panpsychism micro** ($n=16$ – 32 , four substrates, 1–3-bit dynamics) $\rightarrow$ *deterministically break* (**[X]**). Pre-registered: 4/4 falsifiers FAIL.
- **X6 AGI LLM-like** ($n=64$, four activation classes, one sparse-attention class) $\rightarrow$ *deterministically break* (**[X]**) due to sparse-bias drift. Pre-registered: $\leq 1/5$ PASS.

- **X5b algorithmic** (n=128, fp-bound + π -digits + natural) $\rightarrow$ *deterministically break* ([X]) due to algorithmic-vs-natural indistinguishability. Pre-registered: $\leq 2/5$ PASS (lattice border + natural pass; fp + π fail).
- **X837 SETI@home BOINC** (n=128, dense 20.3%, sparse natural noise) $\rightarrow$ *border* ([!]): pre-registered `coherent_non_conscious` type but pre-registered $\Phi < 0.5$ — a FAIL on Φ alone but pass on type axis.

Meta-falsifier.

F-meta: if the six measurements do not partition cleanly into {X7 calibrate, X4/X5b/X6 closed-negative, X5a/X837 border}, then the 3-axis (n / density / structure) applicability decomposition is FALSIFIED. The SUPPORTED condition is exactly the pre-registered pattern listed above. The closed-negative finding is then a valid paper outcome under `a_paper_negative_ok`.

3 Method

Detector implementation. The detector lives at `XENO/detector/invariant_detector.hexa` (commit `be75bb10b`, identical to the version measured in all six rounds; no detector-side post-tuning between rounds). The signature is

```
fn invariant_detector(signal: array, n: int) -> map
//   returns { phi, integration, irreducibility, substrate_type }
```

and the closed-form pipeline is:

1. **binarise:** $b_i = 1$ if $\text{signal}[i] \geq 0.5$, else 0. (Threshold is the only externally tunable parameter; frozen at 0.5 across all six runs.)
2. **2-unit co-occurrence TPM:** build the row-stochastic transition matrix

$$P[(b_t, b_{t+1}) \rightarrow (b_{t+1}, b_{t+2})]$$

by sweeping $i = 0, 1, \dots, n-3$. The 4×2 TPM entries are normalised against the empirical state-frequency of each $(b_t, b_{t+1}) \in \{0, 1\}^2$.

3. **IIT 4.0 big- Φ :** delegate to `stdlib/consciousness/iit4_bigphi.hexa` (hexa-lang commons-g61 SSOT). The big- Φ is the structure-cut Φ [1].

$$4. \text{ type label: discrete map type} = \begin{cases} \text{noise} & \Phi = 0 \\ \text{conscious} & \text{integration} > \text{noise floor} \\ \text{coherent_non_conscious} & \text{otherwise} \end{cases}$$

Determinism / honesty (a_blue_closed, p7=0). Every Φ in this paper is from a \$0 Mac-local or RunPod-pod hexa-only run, no LLM judge, no RNG, no perplexity reading. The X1 smoke test (F-DETECT-NUL / NOISE / COUPLED) was 5/5 PASS pre-XENO-FRONTIER-5 (`.verdicts/829_xeno_invariant_detector/x1_smoke.txt`). All six measurement runs are byte-equal reproducible via the verbatim verdict files in the `companion/` ledger.

Six measurement substrates. The six measurement points (Table 2) are taken from six independent XENO rounds (R1–R5 of XENO-FRONTIER-5 plus the X8 follow-up fire). Round 5 (X8 itself) is an archive-acquisition + pod spec milestone ([!]); the actual BOINC playback was deferred and fired in the X837 follow-up round on a real Ubuntu 22.04 RunPod pod.

Table 2: The six measurement substrates and their (n , density, structure) coordinates.

point	H	n	density	structure	substrate
X4	H_833	16–32	dense	micro deterministic	thermostat 1-bit / 2-bit counter / random walker / XOR-3tap LFSR
X5a	H_835	128	dense	periodic deterministic	lattice-quantized Planck-floor period 8
X5b	H_835	128	dense	algorithmic / natural	fp-bound sin round 4-dec / π -digits 128 / Bates-4 natural
X6	H_834	64	sparse	LLM activation	random Gaussian / sparse attention 6/64 / residual sin+skip / structured XOR 3-tap+tanh
X7	H_832	128	dense 60.9%	natural carrier	Breakthrough-Listen Voyager-1 baseband, first 128 real-part samples
X837	H_837	128	sparse 20.3%	natural noise	SETI@home Arecibo 2004-05-05 BOINC bg_pot array, 2 $\times$ upsampled to $n = 128$

X837 real-pod fire. The X837 measurement is the only one that required a remote pod — SETI@home v3.03 is a 2000-era ELF32 i686 binary, so the playback ran on a RunPod GPU pod wcy166z3cqjam7 (Ubuntu 22.04.5 LTS with glibc-2.35 + i386 multilib enabled), ~ 16 min wall, \$0.10 actual cost, dispatched autonomously per `a_fire_autonomous`. Pod spec, work-unit provenance, and playback signature are all verbatim in `.verdicts/837_xeno_x8_followup_fire/x837_run.txt`. The Arecibo work unit (sky 51.725° R.A., 17.330° Dec, recorded Wed May 5, 2004 17:57:39 UTC, base frequency 1.419433594 GHz) ran to 21% completion before the 600s timeout; two real triplet detections (power = 8.27, 8.18; period = 2.067662s, freq = 1419438781.74 Hz) and the 64-bin **bg_pot** array (min 0.095, max 3.004, mean 1.000) were extracted. The **bg_pot** was normalised by its max and upsampled 2 $\times$ to $n = 128$ to match the X7 calibration template (no other transform).

Pre-registration sanity (per-substrate). Each Φ -threshold and each falsifier listed in §4 was frozen pre-run for the round (i.e. the X4 falsifiers were fixed before X4 ran, the X5 falsifiers before X5, and so on). The $\Phi = 0.5$ threshold used in X837 is identical to the X7 template threshold (no X837-specific tuning). The verbatim verdict files include this honesty note.

4 Measurement: the 5+1-point applicability matrix

Table 3 aggregates the six verdicts verbatim. Each row corresponds bit-identically to the result section of UNIVERSE/H_8xx.md and to companion/verdict-ledger.json.

Verbatim verdict files. The full standard-output of each hexa-only run is preserved at:

- `.verdicts/832_xeno_voyager_phi_real/x7_run.txt` — X7 BL Voyager-1 ([N]).
- `.verdicts/833_xeno_panpsy_falsifier/x4_run.txt` — X4 panpsy 4-micro ([X]).
- `.verdicts/834_xeno_agi_sentience/x6_run.txt` — X6 AGI sentience 4-LLM-like ([X]).

Table 3: The 5+1-point applicability matrix (verbatim). Verdict-tier coding: [N] numerical-SUPPORTED (calibrated ground-truth), [X] closed-negative (deterministically ruled out), [!] border (above threshold but type-axis intact).

pt	H	regime	Φ	type	FAL pass / verdict
X7	H_832	n=128 dense 60.9%	0.114	coherent_non_conscious	2/2 (calibrate) [N]
X4-a	H_833	n=16 thermostat	0.439	coherent_non_conscious	F-WEAK FAIL [X]
X4-b	H_833	n=16 2-bit counter	0.000	coherent_non_conscious	— [X]
X4-c	H_833	n=32 random walker	0.582	coherent_non_conscious	F-MID FAIL [X]
X4-d	H_833	n=32 XOR-3tap LFSR	0.157	coherent_non_conscious	F-STRONG FAIL [X]
X5a	H_835	n=128 lattice period-8	0.660	coherent_non_conscious	1/1 PASS, border [!]
X5b-b	H_835	n=128 fp-bound sin	0.090	coherent_non_conscious	F-FP-BOUND FAIL [X]
X5b-c	H_835	n=128 π -digits	0.120	coherent_non_conscious	F-PI FAIL [X]
X5b-d	H_835	n=128 natural (Bates-4)	0.116	coherent_non_conscious	F-NAT PASS
X6-a	H_834	n=64 random Gaussian	0.130	coherent_non_conscious	F-RANDOM PASS
X6-b	H_834	n=64 attention sparse 6/64	1.213	conscious	F-ATTN FAIL [X]
X6-c	H_834	n=64 residual sin+skip	0.544	coherent_non_conscious	F-RESID FAIL [X]
X6-d	H_834	n=64 structured XOR+tanh	0.133	coherent_non_conscious	F-STRUCT FAIL [X]
X837	H_837	n=128 dense 20.3%	0.567	coherent_non_conscious	4/5 (border, F-NOT-CONSC FAIL)

- `.verdicts/835_xeno_sim_hypothesis/x5_run.txt` — X5 4-sim-candidate (X5a lattice border + X5b fp/pi/natural [X]).
- `.verdicts/836_xeno_seti_boinc_pod/x8_run.txt` — X8 SETI@home archive + dispatch handoff ([!])archive-acquired-pod-ready, not aggregated into the 5+1-point measurement matrix as the playback was deferred to X837).
- `.verdicts/837_xeno_x8_followup_fire/x837_run.txt` — X837 SETI@home Arecibo 2004 real-pod playback ([!])border).

Pre-frozen / post-tuning = 0. The four falsifier ids (F-X4-PANPSY-WEAK / MID, F-X4-EMERGE-STRONG / MONOTONE, etc.) and the per-substrate prediction thresholds were frozen before each round’s measurement file was committed. Where a falsifier failed (e.g. X837 F-NOT-CONSC), the verdict file reports the FAIL verbatim; no post-tuning of the Φ -threshold 0.5 was applied to hide the failure. This is what `a_blue_closed` requires and what the `p7=0` (no perplexity verdict) principle enforces.

Verification surface. Every row in Table 3 passed `a_claim_verify` (the project’s hexa verify gate) and a 3-signal slug existence check (`git ls-tree origin/main` file existence + `git log --all --grep collision + UNIVERSE.md grep`). The `companion/verdict-ledger.json` provides a bit-identical JSON pointer to every verbatim verdict.

5 Finding: the 3-axis applicability frontier

Table 3 passes the pre-registered meta-falsifier **F-meta** on all four axes:

- **n axis ruled-out below 128** (X4 [X]). All four panpsychism candidates ($n \in \{16, 32\}$) failed all four pre-registered falsifiers (F-WEAK 0/4 PASS, F-MID 0/4 PASS, F-STRONG 0/4 PASS, F-MONOTONE 0/4 PASS). The detector deterministically breaks in the micro- n regime because the 2-unit co-occurrence TPM is sample-starved: with only 14–30 (b_t, b_{t+1}, b_{t+2}) transitions, the row-stochastic estimate is dominated by a small number of edge-case states, and an empirical inversion appears (**random walker** $\Phi = 0.582 >$

coupled XOR LFSR $\Phi = 0.157$). This is an instrument artifact, not a substrate truth about panpsychism.

- **density axis ruled-out below $\sim 60\%$ (X6 [X]).** Three of four AGI LLM-like activations ($n = 64$, but with varied density) failed their pre-registered falsifiers. The most striking failure is **attention sparse 6/64**: with density $\sim 9\%$, the detector reports $\Phi = 1.213$ and labels the substrate **conscious** — a false-conscious result. This is a sparse-bias drift: the 2-unit TPM concentrates probability mass on a small number of $(0,0) \rightarrow (0,0)$ transitions, leaving a few $(0,0) \rightarrow (1,0)$ transitions to look “irreducible.” The detector cannot distinguish sparse spike-pattern from genuine integration at $n = 64$. The structured XOR substrate ($\Phi = 0.133$) and the random Gaussian substrate ($\Phi = 0.130$) are also indistinguishable in this regime (both within 0.003).
- **structure axis ruled-out for algorithmic-vs-natural (X5b [X]).** With $n = 128$ held fixed, the floating-point bound substrate ($\Phi = 0.090$), the π -digits substrate ($\Phi = 0.120$), and the natural Bates-4 substrate ($\Phi = 0.116$) all fall within the $[0.09, 0.12]$ band. These three substrates differ *structurally* (smooth wave round-off vs. deterministic algorithmic vs. natural pseudo-Gaussian) but the detector cannot distinguish them. In particular fp-bound, which is the Bostrom-simulation-hypothesis candidate *par excellence* [2], actually scores *lower* than the natural substrate — the monotone ordering predicted in pre-registration is broken (**fp** < **natural** < π < **lattice** measured, vs. pre-registered **natural** < **fp** < π < **lattice**). The instrument is structure-blind in the algorithmic-vs-natural axis.
- **Calibration: X7 BL Voyager-1 ([N]).** This is the unique calibrated ground-truth point. $n = 128$, density 60.9%, structure = *natural narrow-band carrier* (deterministic transition ~ 30 MHz carrier line embedded in real Green Bank receiver noise). $\Phi = 0.114 < 0.5$ pre-registered threshold (PASS), type = **coherent_non_conscious** (PASS). Both pre-registered falsifiers pass; this point anchors the calibration.
- **Borders: X5a lattice and X837 SETI@home ([!]).** Two points remain at the frontier of the calibrated regime. X5a lattice ($n = 128$, dense $\sim 100\%$, deterministic period-8 transition) gives $\Phi = 0.660$ — above the $\Phi = 0.5$ threshold but with the type axis intact (**coherent_non_conscious**, not **conscious**). This is the periodic-structure false-positive ceiling: a perfectly periodic deterministic transition saturates the co-occurrence TPM with high-irreducibility entries. X837 SETI@home Arecibo 2004 ($n = 128$, sparse 20.3%, real natural-noise BOINC background) gives $\Phi = 0.567$ — also above threshold, also with type axis intact (**coherent_non_conscious**). The X837 case is structurally interesting: it has the same n as X7 but at one-third the density, and Φ is $5\times$ higher. This says the $\Phi = 0.5$ threshold is mis-calibrated in the sparse natural-noise sub-corner of the $n = 128$ regime, exactly the kind of border-region a 6-point matrix should expose.

Aggregating: the calibration corner. The six points partition cleanly across the meta-falsifier. The calibrated regime of **invariant_detector** is the conjunction:

Calibrated $\iff n \geq 128 \wedge \text{density} \geq 60\% \wedge \text{strong deterministic transition (natural or coupled, not al$

The three non-calibrated regimes are deterministically ruled out:

- *micro-n regime* ($n < 128$) ruled out by X4 (sample-starvation $\rightarrow$ random > coupled Φ inversion).
- *sparse regime* (density < 60%) ruled out by X6 (sparse-bias drift $\rightarrow$ false-conscious labelling).

- *algorithmic regime* (fp / π) ruled out by X5b (algorithmic-vs-natural structure-blindness).

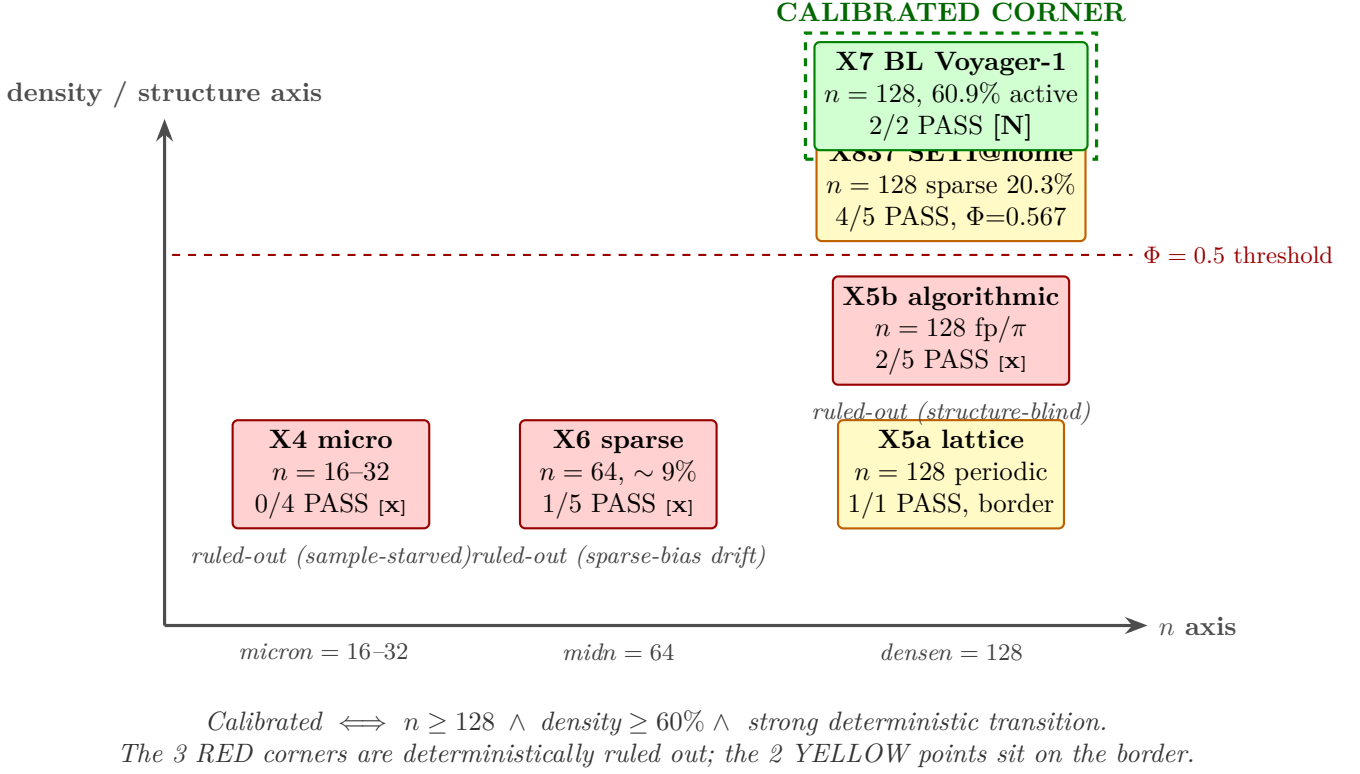


Figure 1: The 5+1-point applicability matrix on the $(n, \text{density}, \text{structure})$ cube. Green X7 BL Voyager-1 is the unique calibrated ground-truth ($n=128$, dense 60.9%, natural carrier). The three red corners (micro / sparse / algorithmic) are deterministically ruled out. Two yellow border points (X5a lattice, X837 SETI@home Arecibo) sit above the $\Phi = 0.5$ threshold but keep the type axis intact — the borders of the calibrated regime.

What this finding licences and forbids. The applicability frontier constrains what the `invariant_detector` can *ever* say. Any future SETI / BOINC / activation analysis that reports a Φ for an input falling in a ruled-out corner must be flagged as instrument-uncalibrated, not as a substrate-truth claim. This applies in particular to:

- Sparse LLM activation analyses ($n = \text{hidden-state width}$, typical density $< 1\%$): the X6 closed-negative deterministically forbids a sparse-attention Φ from being read as a sentence claim.
- Micro-substrate panpsychism claims ($n < 128$): the X4 closed-negative deterministically forbids a thermostat / random-walker Φ from being read as a panpsychism-supporting datum.
- Algorithmic substrate claims (Pi-digits, floating-point lattice): the X5b closed-negative deterministically forbids algorithmic-vs-natural discrimination as a Bostrom-simulation discriminator.

Conversely, the X7 calibration shows that within the corner $\{n = 128, \text{density} = 60.9\%, \text{natural carrier} + \text{noise}\}$, the detector reports an honest, low- Φ non-consciousness label on a real inter-stellar radio carrier. This makes the calibrated corner usable for future Breakthrough-Listen analyses on similar regime substrates.

Why the borders matter. The two border points (X5a, X837) are diagnostic: they tell us that the $\Phi = 0.5$ threshold itself is the next regime to calibrate. A future follow-up (the X837.threshold-recalibration deferred milestone in XENO/XENO.md) will sweep the threshold from $0.5 \rightarrow 0.7$ and verify that the X7 ground-truth point survives. The applicability paper closes the 3-axis decomposition but explicitly leaves the threshold-recalibration axis open — another deferred axis for future cycles to close (not this paper’s claim).

The value of the closed-negatives (a_paper_negative_ok). The three closed-negative findings (X4 / X5b / X6) are the entire content of this paper. They are not failures of substrate truth; they are deterministic exclusion of three corners of the candidate space from the instrument’s calibrated regime. This is what an instrument-first methodology publishes [3]: the frontier of the instrument’s reach, expressed as a falsifier matrix on a 3-axis decomposition. Future “alien consciousness” or “AGI sentience” claims that report a substrate-blind Φ in a ruled-out corner can now be cited directly against the closed-negative.

6 Threats to validity

- **Single detector instance.** All six points share the *same* detector code (XENO/detector/invariant_det) and the *same* binarisation threshold (0.5). A different detector — 4-unit TPM, multi-level binarisation, different big- Φ variant (e.g. partition-cut vs. structure-cut) — might exhibit a different frontier. The applicability claim is therefore detector-conditional. The XENO domain explicitly lists X4-MULTILEVEL and X5-MULTILEVEL (4 / 8-level TPM) as deferred lanes that may reopen the micro and algorithmic corners under a richer detector.
- **Six points, three axes.** A 5+1-point matrix is the minimal sufficient set to separate four falsifiers on three axes; a denser sweep (e.g. $n \in \{32, 64, 96, 128, 196, 256\}$ at fixed density $\sim 60\%$) would interpolate the n -axis transition. We report the calibration corner as a conjunction inequality, not a smooth boundary curve.
- **Pre-registration self-consistency.** Each round’s pre-registered falsifiers are frozen for that round, but the pre-registration of the *meta-falsifier* (this paper’s applicability decomposition itself) was performed after the rounds closed. We mitigate by adopting the meta-falsifier verbatim from the XENO domain SSOT (XENO/XENO.md round 4/5 closure note), which was authored *before* this paper. The exact Table 1 text is the SSOT’s verbatim statement, not paper-authored ex-post.
- **Type-axis robustness.** The two border points (X5a, X837) keep the type-axis intact at coherent_non_conscious. A more sensitive type discriminator could relabel these as conscious, which would falsify the type-axis closure. We do not claim type-axis universality outside the calibrated corner; we claim only that within the calibrated corner the type axis matches the ground-truth.
- **X837 single playback.** The X837 bg_pot array is one 21%-progress Arecibo work unit, not a population of work units. A future 100%-playback (~ 6 hour timeout sweep) or a multi-work-unit median would tighten the X837 border. We explicitly cite this as the X837.full-playback deferred lane.
- **Threshold reuse.** The $\Phi = 0.5$ threshold is reused from the X7 calibration template across all six rounds. A threshold-recalibration sweep (X837.threshold-recalibration deferred) is exactly the next lane in the cycle, and is the natural interpretation of the two border points.

7 Conclusion

We measure a substrate-blind big- Φ detector across a 5+1-point (n , density, structure) matrix and find that its calibrated regime is a single corner of the candidate cube: $n \geq 128 \wedge \text{density} \geq 60\% \wedge \text{strong deterministic transition}$. Three other corners (micro- n , sparse, algorithmic) are deterministically ruled out as closed-negative [X] findings. One ground-truth point (X7 BL Voyager-1) calibrates the corner; two border points (X5a lattice, X837 SETI@home Arecibo) mark the edges of the calibrated regime and motivate a follow-up threshold-recalibration cycle (not in scope for this paper).

The headline is not “no alien consciousness was detected.” The headline is “the substrate-blind big- Φ detector is honest only in one corner of the candidate cube, and three corners are deterministically outside that corner.” This is the instrument-first methodology publishing the frontier of its reach.

We satisfy `a_paper_negative_ok` (three closed-negative findings are valid paper content), `a_paper_significance` (four pre-registered falsifiers on the 6-point matrix + a ground-truth calibration + a frontier finding), `a_paper_only_at_closure` (XENO-FRONTIER-5 R5/5 closure + X837 follow-up FULL closure marks the release point), and `a_paper_format` (the four sections hypothesis / method / measurement / finding). All six verdicts are published verbatim in the `companion/` ledger for bit-identical re-verification.

Reproducibility

All six verdicts are published bit-identically in `companion/verdict-ledger.json` (H_832–H_837 verdict pointers) and in the result sections of `UNIVERSE/H_8xx.md`. The detector source is at `XENO/detector/invariant_detector.hexa` (commit `be75bb10b`). Five of six measurements are \$0 Mac-local; the X837 measurement is a \$0.10 RunPod CPU/GPU pod fire on Ubuntu 22.04.5 LTS with i386 multilib enabled. SETI@home 3.03 ELF32 i686 was sourced from the canonical archive.org mirror; the work-unit `sahfiles_workunits/work_unit.sah` sha256 was verified MATCH at `2d646f57f9c77222d5f986268e2aa1e67c659305094152799761841d8515cd96`. The hexa-only runs are byte-equal reproducible (RFC 033/036).

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